

Adoption and Market Trends of Electric Vehicles in Europe

I. Introduction

A. Motivation and context

1. Personal and professional link to EV topic

I've been fascinated by cars since I was a kid. That interest has stayed with me and eventually led me to Škoda Auto University. During my studies, I had the chance to complete a long-term internship in the IT Development department at Škoda Auto, where I worked with vehicle data and digitalization projects, like enterprise data architecture. Through this activity at Škoda Auto, I became especially interested in how data and technology can help shape future of automotive companies and the field of mobility as a whole.

The EV topic connects both my personal passion for the automotive world and my professional and academic focus on data analytics. That's exactly why I chose this as the subject of my final project.

2. Current state of EV adoption in Europe and why it matters

The shift to electric vehicles is the biggest transformation the car industry has seen in decades. But while some countries in Europe are pushing ahead quickly, others are falling behind

Rather than relying on assumptions or intuition, I think it's crucial to look at real data, because the pace and pattern of EV adoption in Europe will play a major role in achieving climate targets and shaping the future of transportation.

B. Research focus

1. Three main drivers of EV adoption: Economy, Emissions, Infrastructure

The goal I set for this project was to explore what drives the adoption of electric vehicles across Europe. After considering several possible angles, I decided to focus on three key areas: economy related factors (clean energy investments, GDP), emissions (share of clean energy, transportation related emissions), and charging infrastructure. There are many other factors that come into play; but for the purpose of this project, these three are ideal.

Economic strength affects how affordable EVs are for consumers and how much governments can invest in incentives. Emissions reflect the environmental urgency and political pressure to lower greenhouse gasses output, which directly influence personal mobility. Infrastructure, meanwhile, plays a practical role. It determines whether driving and charging an EV is feasible

for most people. By analyzing these three areas together, I believe I'll get a complex view of what really matters for EV adoption in different countries.

II. Methodology

A. Data Overview

1. Sources, datasets and variables

For this project, I worked with a total of eight datasets from publicly available European and global sources. I used sources like European Alternative Fuels Observatory (EAFO), Eurostat, the IEA, and the EEA.

The datasets that I'll use in the analysis are:

- BEV market share (AF Market Share of Total Registrations, EAFO)
- GDP per capita in PPS (Eurostat)
- BEV total numbers (EAFO)
- EV sales and market share over time (IEA Global EV Data)
- Recharging point types and counts (EAFO)
- Transport-related emissions (National Emissions Database, EEA)
- Clean electricity production and consumption (Monthly Electricity Statistics, IEA)
- Clean energy investment (World Energy Investment 2024, IEA)

Two variables, *Country* and *Year*, were present across most datasets, which allowed for merging and time-series analysis. In the datasets with a date in a different format the year was extracted using *input* and *scan* operations.

In contrast, some datasets contained specific variables related to the dataset's main focus, such as *Pollutant_name*, *Balance*, *Fast_DC*, *Product*, *Powertrain* and many more.

2. Missing sources and limitations

Some countries had missing values, most notably Bulgaria and Croatia, for which final electricity consumption data was unavailable. This made clean electricity share impossible to compute for those cases. Additionally, land area data had to be manually written into the code since it wasn't available in any of the used datasets.

Although other socio-political, economical and behavioral variables (e.g. incentives, car ownership norms, price and availability of certain BEV models or whole makes, resistance to change across Europe) have a significant influence on the BEV adoption as well, they were not considered in this project, since I couldn't find any comprehensive public dataset containing the relevant information. Still, I believe that multiple parts of my project can provide an interesting and valuable insight into this problematic.

B. Analytical approach

1. Introduction to SAS and relevant analytical tools

All datasets in this project were processed, cleaned, and analyzed using SAS. Since I was focusing mostly on the relationship between multiple observations, I mainly used statistical tools like:

- *proc corr*
- *proc reg*
- *proc standard*

Other tools that I've used include:

- *proc import*
- *proc sql*
- *proc print*
- *proc sort*
- *proc gplot*, *proc sgplot*
- *proc gchart*
- *proc gmap*

2. Merging, standardizing, and calculating new variables

After importing the datasets, I began by harmonizing and cleaning them to ensure they could be used together. The first step involved filtering the datasets to include only the 28 European countries selected for analysis. This was done using an if/then filter based on country names. To make merging easier later, I also renamed columns with consistent names. Standardizing the column that contained the country name across all datasets was essential.

Some datasets required more specific manual adjustments. In the Monthly Electricity Statistics, the first 8 rows had to be removed because they did not contain usable data. Column names were not automatically readable in SAS, so I had to manually rename them. I encountered a similar issue with the Charging Points dataset, where charger type names were long, so I simplified and renamed them by hand. Missing charger counts were replaced with zeros using the *coalesce(VAR, 0)* function to avoid gaps during calculations.

The clean energy investment dataset (Excel format) presented a challenge because the years were represented as column headers and not as a tidy time series. To overcome this, I used a workaround that manually assigned the investment values to their respective years (from y_0 to y_0+i) during the cleaning step.

Once cleaned and aligned, the datasets were merged, in each part and operation separately, using *Country* and *Year* as keys. I then created several calculated variables, including:

- Clean electricity share
- Charging Points Index

- Charging Index per 1000 km²
- BEVs per charger
- Index per 1000 BEVs
- Aggregated values for Europe-wide emissions and BEV sales
- Composite score

For the infrastructure-related analysis, I constructed a custom Charging Points Index, which assigns weights to each charger type depending on its speed and charging power. This helped reflect not just quantity, but also the quality of public charging infrastructure. In addition, I used the manually defined land area values to standardize infrastructure density by country size.

To combine variables of different scales (percentages, indices, density), I applied *proc standard* to convert them into z-scores. These standardized values were then used to calculate a composite score, which served as a central indicator in the final comparison of EV ecosystems.

III. Analysis and Results

A. Economic and Environmental Drivers of BEV Adoption

This part looks at how economic performance, clean electricity, and transport emissions affect the market share of battery electric vehicles (BEVs) in Europe from 2013 to 2023.

1. BEV Market Share and GDP per Capita

The first step was to check whether richer countries tend to adopt more BEVs. I used GDP per capita in PPS (Purchasing Power Standards) and compared it to BEV market share across 28 countries. The correlation result was $r = 0.322$, and the relationship was statistically significant ($p < 0.0001$). This confirms a moderate positive connection between income levels and BEV market share.

To explore this further, I ran a simple linear regression. The result showed a significant effect of GDP on BEV adoption:

- $\beta = 0.12707$, meaning that for every 1-point increase in *GDP* index (where EU average = 100), *BEV market share* increases by around 0.13%.
- The model was statistically strong ($F = 52.85$, $p < 0.0001$) but had limited explanatory power ($R^2 = 0.1037$). This means *GDP* alone explains only about 10% of the variation in *BEV market share*. This low-to-moderate explanatory power is normal for socio-economic data, where many other factors also play a role.

2. Clean Electricity and Transport Emissions

To build a better model, I added two more variables: *clean electricity shares* and *transport-related CO₂ emissions*.

The share of green electricity was calculated as:

$$\frac{\sum \text{Net Electricity Production from (Total Renewables (Hydro, Geo, Solar, Wind, Other) + Nuclear)}}{\sum \text{Final Consumption (Calculated)}} \times 100$$

There were missing values for Bulgaria and Croatia due to unavailable electricity consumption data.

The improved model used 432 observations and explained more variance ($R^2 = 0.1793$). All three variables were statistically significant:

- *GDP per capita*: $\beta = 0.10051$, $p < .0001$
- *Clean electricity share*: $\beta = 0.10806$, $p < .0001$
- *Transport emissions*: $\beta = -0.00002053$, $p = 0.0012$

This confirms that BEV adoption increases with both higher income and greener electricity. The small negative effect of emissions likely means that countries with high transport emissions are

either not promoting EVs effectively or are experiencing a delayed policy response. In any case, high emissions alone do not automatically result in higher BEV uptake.

A matrix plot visualized the relationships and confirmed the trends: *GDP* and *clean electricity* had positive effects on *BEV market share*, while *transport emissions* had a slight negative correlation.

B. Charging Infrastructure and EV Adoption

This part looks at how the availability and quality of public charging infrastructure relate to BEV uptake across European countries.

1. Introduction of Charging Points Index

To measure charging infrastructure quality, I introduced a custom indicator called the *Charging Points Index*. It assigns weights to each type of charger based on speed and power:

- Slow AC (<7.4kW) = 1
- Medium AC (7.4–22kW) = 2
- Fast AC (>22kW) = 3
- Slow DC (<50kW) = 4
- Fast DC (50–150kW) = 6
- Ultra-fast DC L1 (150–350kW) = 8
- Ultra-fast DC L2 ($\geq$ 350kW) = 10

This way, countries with fewer but higher-capacity chargers are not undervalued. The index helps show not just how many chargers there are, but how “good” or fast they are. In addition to that I have also calculated the total number of chargers.

2. BEV-to-Charger Ratios and Infrastructure Adequacy

To make the SAS maps dataset accept my list of countries, I had to harmonize country names with the map dataset (I found out the names using *proc datasets lib=mapsgfk* command). Then I also defined the color scale for both maps.

Two maps were created to visualize infrastructure adequacy:

- Map 1 (*BEVs per charger*):
 - o Green = fewer BEVs per charger, therefore better access to recharging.
 - o Red = high BEV-to-charger ratio, which may result in a possible congestion risk.
- Map 2 (*Charging Index per 1000 BEVs*):
 - o Green = Higher index and better-quality infrastructure per BEV.
 - o Red = Lower index and a worse-quality infrastructure per BEV.

It is important to mention, that even though some countries are red or orange on these maps, this doesn't necessarily mean bad charging infrastructure. Norway, for example, is known for its high BEV market share and a solid charging infrastructure. The market share of BEVs in certain

countries can outweigh the charging points numbers and indexes. The inverse effect then applies to countries like Czech Republic and Croatia, where ICE automobiles are still dominant.

Then I tested whether better infrastructure leads to higher BEV adoption. The correlation between the *Charging Points Index* and *BEV market share* was very weak ($r = 0.194$, $p = 0.3227$) and not statistically significant. The regression confirmed this:

- $R^2 = 0.0376 \rightarrow$ index explains only 3.8% of the BEV share
- $p = 0.3227 \rightarrow$ model is not meaningful

Charging Points Index does not predict *BEV market share* well. Countries with high market share do not necessarily have proportionally better infrastructure quality.

However, there was a strong correlation between *Total Chargers* and *Total BEVs*:

- $r = 0.90$, $p < 0.0001$

Regression:

- $R^2 = 0.6729 \rightarrow$ number of chargers explains 67% of total BEV fleet size
- This relationship is statistically very strong and clearly meaningful

This shows that while infrastructure quality doesn't predict market share well, the quantity of chargers clearly tracks actual BEV adoption. Countries most likely expand infrastructure as demand for charging grows.

C. Impact of Clean Energy Investments (EU-wide)

This section examines how clean energy investment at the EU level affects EV sales and transport-related emissions between 2015 and 2023. All datasets have been aggregated for the whole Europe.

1. Clean energy spending and its influence on EV sales and emissions output

The regression analysis confirms a very strong relationship between *clean energy investment* and *BEV sales*. The model achieves $R^2 = 0.9610$ ($p < 0.0001$), meaning that over 96% of the variation in BEV sales can be explained by changes in investment levels. This is one of the strongest results in the entire project.

The slope coefficient ($\beta = 7210.89$) suggests that every additional billion USD invested in clean energy corresponds to an increase of roughly 7,211 BEVs sold in Europe. The fit plot shows all visible points sitting tightly around regression line. This supports the hypothesis that clean power infrastructure creates favorable conditions for large-scale EV adoption.

The second regression, however, is surprisingly to me on the other end of the spectrum. It tested whether the same investment explains *transport-related CO₂ emissions*. The results are following: $R^2 = 0.0582$, $p = 0.5316$, showing no statistically significant relationship. Although emissions slightly declined around 2021–2022, this cannot confidently be attributed to the

investment increase. The observed decline can rather be attributed to pandemic-related mobility restrictions during Covid-19.

Several factors may explain the weak relationship between *clean energy investment* and *transport-related CO₂ emissions*:

- Investment effects may take longer to materialize, especially in emission statistics
- Growing demand for transport may be outweighing the decarbonization efforts in the transportation sector
- Emissions are influenced by many other sectors (goods transport, flights, and ship transport), not just passenger road transport

The line plots help visualize both dynamics. Investment and BEV sales follow a clear synchronized upward trend, especially from 2020 onward. Emissions, on the other hand, remain relatively flat, even slightly rebounding in the last years.

D. Most Balanced EV Ecosystems in Europe

This final part brings together the three of the key pillars of a successful EV ecosystem: market adoption, infrastructure availability, and clean electricity. Instead of looking at each of them in isolation, I created a unified metric that allows for a fair cross-country comparison. The goal was to identify which countries offer the most balanced environment for electric mobility, and where further improvements are still needed.

1. Standardizing Key Variables for Comparability

To compare BEV adoption, charging infrastructure, and clean electricity across countries, I had to standardize the data. I used *proc standard*, which was new to me, to transform all three variables into z-scores. This allows the different indicators, in my case percentages, indexes, and density per km², to be combined and compared meaningfully. Before standardizing, I adjusted the *Charging Points Index by Land Area* (per 1000 km²), since absolute number of charging points varies widely depending on country size. Since I forgot to include land areas of European countries as one of my datasets, I had to write the values directly in code to avoid working with another external file. The values were sourced from Eurostat.

2. Composite Score Calculation and Country Rankings

The composite score was created by taking the mean of the three standardized indicators. This approach gives equal weight to *BEV market share*, *charging infrastructure density*, and *clean electricity share*, displaying EV ecosystem maturity in a single metric. Based on this score, countries were sorted and grouped into three readiness ranks (*High, Medium, Low*) and further into nine color groups for the map. These visual clues make it easier for the reader to determine where each country stands in the rankings.

3. Interpretation of Results and Examples

The results show huge differences across Europe. Some countries score well across all three dimensions, but many are surprisingly uneven in the three metrics, e.g. dominant clean electricity share without a significant EV uptake or well-developed charging infrastructure without adequate EV adoption. The composite scoring reveals that clean energy, charging infrastructure, and an active market presence all matter.

In addition to the table with all the scores I also produced a bar chart and a map. Both clearly highlight where countries are leading, catching up, or lagging behind. In my opinion, these visuals can serve as a general overview for policymakers and provide a basis for future policy decisions or investment focus, especially in countries that are strong in one area but weak in another.

IV. Conclusion

A. Summary of Key Findings

This project explored the key drivers behind battery electric vehicle (BEV) adoption across Europe, using a combination of economic, environmental, and infrastructure-related indicators. The analysis revealed several clear trends. GDP per capita and clean electricity share were both positively associated with BEV market share, while transportation-related emissions showed a small but significant negative relationship. Infrastructure quality (as captured by the *Charging Points Index*) did not predict adoption well, but the number of public chargers strongly correlated with the total number of BEVs.

The strongest statistical result came from the EU-wide regression of clean energy investment on BEV sales, with an R^2 of over 0.96. This clearly illustrates the critical role that long-term infrastructure investments play in market expansion. However, the same investment had little to no measurable effect on transport-related emissions; at least not within the studied time period. This suggests that either the effects are delayed or that investment alone is not sufficient to reduce emissions in a sector with many other influencing factors.

B. Strategic and Stakeholder Implications

1. Implications for Policy Makers

These findings underline the need for a holistic policy approach. If the goal is not only to grow the EV market but also to cut emissions, then clean energy investment must be supported by targeted policies as well. These may include measures like fuel taxation, support for low-emission logistics, incentives for public transport, or modal shift initiatives such as moving cargo from roads to rail.

2. Automakers' Perspective

From the automaker's perspective, the results suggest potential in countries that are structurally well-prepared, thanks to generating a major part of electricity from green energy sources or having a relatively developed charging infrastructure, but where BEV adoption remains low. These markets may offer growth opportunities, especially if supported by the right incentives or localized product strategies. Automakers could benefit from proactively entering these under-saturated but EV-ready environments.

3. Consumer-Centric View

For consumers, the decision to buy an EV depends not only on price or policy incentives but also on practical and environmental considerations. In countries where charging stations are sparse and slow, driving an EV may not be practical. Likewise, if the electricity mix relies heavily on fossil fuels, the climate benefit of switching to EVs is significantly reduced.

This underscores the need for better alignment between charging infrastructure, energy policy, and environmental goals, so that switching to an EV is not only convenient for consumers, but also represents a meaningful positive change for the climate.

C. Data Limitations and Future Analysis Potential

While the analysis included a broad range of datasets, there are still several data sources that would help deepen the understanding of what drives BEV adoption. It would be valuable to have access to reliable, cross-country data on EV purchase prices, government incentives (such as subsidies, tax benefits, or toll exemptions), and charging costs.

Detailed information on local policies, urbanization levels, battery lifecycle data or average driving distances could also add missing context. These are all factors that are likely to have a significant impact on people's willingness to switch to an electric car. Finally, consistent energy statistics for all countries, including currently missing consumption data for Bulgaria and Croatia, would improve the accuracy of electricity-related indicators.